

Rahul Anand Nayanegali

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SUMMARY

Software Engineer with 6 years of experience building full-stack and backend systems across Amazon Seller Central, AI-powered data tools, and high-traffic streaming platforms. Strong in Java, TypeScript, Node.js, Spring Boot, PostgreSQL, REST/GraphQL APIs, AWS, and production-quality testing.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript

Backend: Java/Spring Boot, Node.js, Flask, REST APIs, GraphQL, gRPC, WebSockets, SSE

Databases: PostgreSQL, SQL

Cloud/Infra: AWS ECS, EC2, Lambda, S3, CloudFront, Docker, CI/CD

Frontend: React, Next.js, TypeScript, Module Federation, Micro-Frontends

Testing: TDD, BDD, Pact, k6, Playwright, Cypress, Jest, Pytest

AI Engineering / Developer Tools: harness engineering, MCP, Claude Code, Kiro, OpenCode CLI, prompt engineering

EXPERIENCE

Software Development Engineer

Jun 2025 – Dec 2025

Amazon

Seattle, WA

- Co-owned Amazon Seller Central's compliance platform supporting revenue-critical seller listing workflows; improved platform reliability by reworking API contracts, frontend integration paths, and regression testing around compliance flows.
- Designed and launched a **Java REST BFF** service for Amazon Seller Central's compliance platform, aggregating 3 internal APIs into a stable widget contract, centralizing orchestration/error handling, and reducing client round trips from 3 to 1 for revenue-critical seller listing workflows.
- Re-architected ActionWidget from hardcoded regulation-specific UI logic into a schema-driven platform using the Strangler Fig pattern; decoupled regulation logic from rendering so new compliance flows onboard via config alone with no code changes.
- Established **TDD** practices across the team by authoring tests around API contracts, error handling, and edge cases; enforced through **trunk-based development** and code review gates, stabilizing production releases and reducing regression risk on revenue-critical seller flows.
- Designed AI-assisted **Playwright** test authoring pipeline using **BDD** specifications as LLM context, cutting test authoring time 63%, raising E2E coverage 60%, and accelerating delivery velocity 30%.

Software Engineer

Aug 2024 – Jun 2025

Shoptaki

New York, NY

- Built a Node.js backend for a natural-language analytics interface that extracted live PostgreSQL schema metadata, relationships, constraints, and join patterns to generate context-aware SQL from user prompts.
- Added SQL validation, sanitization, and staging-database execution safeguards before query execution, reducing risk from malformed or unsafe LLM-generated SQL.
- Built the query interface in React and TypeScript to handle dynamically shaped result sets where columns and types were unknown at build time; implemented async execution states, error surfaces, and adaptive table rendering to handle unpredictable LLM output shapes.

Software Engineer

Feb 2021 – Jan 2023

CodeCraft Technologies

Bangalore, India

- Co-owned **React/TypeScript** component development for JioCinema on **Next.js 13** (Pages Router) with **Redux**; implemented **SSR** and code splitting via dynamic imports on streaming pages that served IPL and FIFA World Cup 2022 traffic, platform peak of 32M concurrent viewers.
- Built **Node.js** and Python backend services on **AWS ECS/Lambda** to process streaming-platform metadata and content workflows, exposing REST APIs and scheduled jobs for ingestion, transformation, and downstream UI consumption.
- Built full-stack audience-engagement features for Voot, including configurable surveys, show-specific quizzes, participant leaderboards, and real-time voting, deployed across multiple entertainment and reality-show properties on a platform serving 100M monthly active users.
- Modernized build pipeline for a high-traffic streaming platform (Webpack/Babel upgrade, asset optimization); cut build times 50% and unblocked ES6+ feature adoption.
- Migrated application state from **Redux** to **Apollo Client (GraphQL)**, implementing a unified GraphQL infrastructure layer; eliminated boilerplate and reduced bundle size 20%.

Assistant Systems Engineer

Nov 2018 – Apr 2020

Tata Consultancy Services

Bangalore, India

- Built real-time IoT monitoring dashboard using **React**, **Flask**, and **Server-Sent Events (SSE)**, handling continuous sensor data streams at scale; designed RESTful API endpoints for auth, data retrieval, and live updates.

EDUCATION

Pace University

Master of Science in Computer Science

New York, NY

January 2023 – May 2024

Sri Siddhartha Institute of Technology

Bachelor of Engineering in Electronics and Communication Engineering

Tumkur, India

September 2018